Lillian Holl

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Spatio-Temporal Patterns of Specimen Contributions in the Donald W. Davidson Herbarium Indicate Co-Collection Is Important For Overall Collection Growth

## Introduction

An herbarium is a library of pressed and dried plants. A physical collection of specimens is built through the process of collection, where collectors identify plants out in the world to be added (see Figure 1). Typically, any plant can be added to an herbarium; however, unique species, locations, or times of year may add more depth to the overall collection. Collectors record information about the plant, such as species, who is collecting it, when and where it was collected, and the habitat it was collected in, along with other information. The plant is then picked and pressed between newspaper sheets to dry, and an herbarium label is created with the gathered information. The pressed plant and herbarium label are then mounted on an herbarium sheet, and the specimen is accessioned, or formally integrated, into the collection.

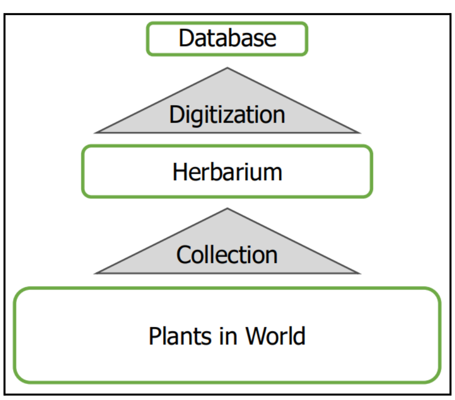


Figure 1 - Conceptual Model of Collection and Digitization Processes.

An additional process called digitization may take place to represent the physical collection virtually in a database (see Figure 1). During digitization, the information from the herbarium label is transcribed and entered into a spreadsheet-like format, where each row corresponds to a specimen. The process may also include taking a high-quality photo of the mounted specimen.

Digitized records of herbarium specimens can be used for scientific studies, but they are not a perfect source of biodiversity information. One reason is that the process of digitization may affect representation in a database[1] because not all physical specimens in incompletely digitized collections may have a corresponding database entry; this is known as digitization bias. A second reason is that biases in the physical collection itself are the result of opportunistic collection by botanists going out when and where they can according to funded projects, time limitations, and personal preference among other reasons. A common example is temporal bias where collection efforts are not consistent over time,[2–4] which leads to specimens either over- or under-representing years in a collection. Biases also exist for the location a specimen was collected; plants are often collected close to an herbarium,[2] and prolific collectors that collect many plants visit so many places that they may end up "clustering" geographically.[3] Individual collectors also vary in how much they personally contribute to an herbarium,[3,5] and it is often a small group of prolific collectors that build the bulk of a collection.[2,3]

The Donald W. Davidson Herbarium, located at the University of Wisconsin - Superior, is a small herbarium with both digitization and collection biases. This collection, registered on the Index Herbariorum with the four-letter code SUWS, contains approximately 10,000 specimens as of February 2025.[6] The only digital records of specimens available are on WisFlora, the Online Virtual Flora of Wisconsin hosted by the Consortium of Wisconsin Herbaria. However, the collection is incompletely digitized as several projects have added specimens to the collection since the last digitization effort in 2007. To facilitate further use of this collection and support future digitization, I aimed to answer several questions about the biases already present in the collection, including who had contributed the most specimens, where they were collecting them, and when they made their contributions to SUWS.

## Methods

For an example of digitization bias, I compared physical specimens and WisFlora records of *Abies balsamea*, the balsam fir. I looked at the herbarium label for the 38 physical specimens present in the collection, and then searched for matches in the 22 WisFlora records. I also categorized where the specimens without WisFlora records were collected, as only specimens collected in Wisconsin are eligible to be in WisFlora.

For the analysis of records that have been digitized, I downloaded all 5,290 records currently available for SUWS on WisFlora[7] and loaded the data into RStudio. I then cleaned the data for easier use during analysis. This process involved removing 56 irrelevant columns and 6 specimens with imprecise geographic information below the state level. Records of 31 specimens were duplicated in the database, and 28 were able to be verified against the physical specimen. The 28 extraneous duplicate records, along with the 6 records that could not be verified, were also removed from the downloaded SUWS dataset. 5,250 records remained for analysis, and additional code in R[8] using the tidyverse[9] and UpSetR[10] packages sorted, counted, and visualized this data. Additionally, ArcGIS Pro was used with wrangled data to create geographic distributions at the Wisconsin county level of the top seven collectors.

## Results

Digitization bias is likely present in the wider SUWS collection, exemplified by records of *Abies balsamea*, the balsam fir (see Figure 2). SUWS has 38 physical *Abies balsamea* specimens; however, only 22 WisFlora digital records exist. 17 physical specimens have a matching database record. This leaves 5 records in the database where no herbarium sheet was found. Of the 21 physical specimens without a digital record, 7 were collected in Wisconsin. While a Wisconsin location typically qualifies a specimen's digital record to be included in WisFlora, these 7 were collected after the last digitization was done in 2007. Physical specimens without database records like these 7 *Abies balsamea* are not present in the following analysis.

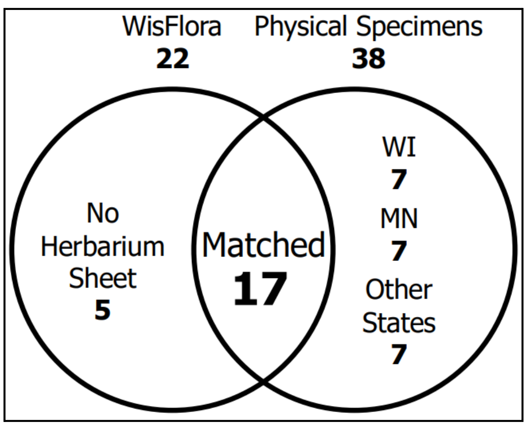


Figure 2 - Digitization Bias Venn Diagram of SUWS *Abies balsamea* Specimens

In the cleaned data, 5,250 specimens represented the contributions made by 292 individual collectors. Collectors contributed 6,485 times in those 5,250 specimens; the contribution number is higher than the specimen number because multiple collectors can be listed as contributing the specimen. A large share of contributions were made by prolific collectors; seven collectors who contributed more than 125 individual specimens make up 49% (3,195) of the contributions to the collection (see Figure 3). These seven are Brashier with 850 contributions; John W. Thompson, Jr. with 616; Derek S. Anderson with 508; Rudy G. Koch with 390; R. Castle with 343; Romans with 321; and Donald W. Davidson, whom the collection is named after, with 168 (See Figure 4).

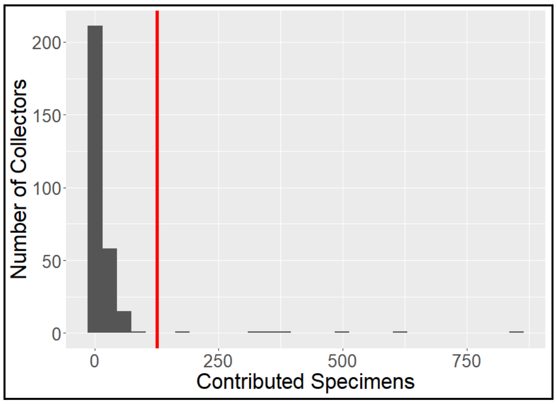


Figure 3 - A Comparison Of Specimens Contributed By An Individual Collector Versus The Number of Collectors. The red line represents 125 contributed specimens.

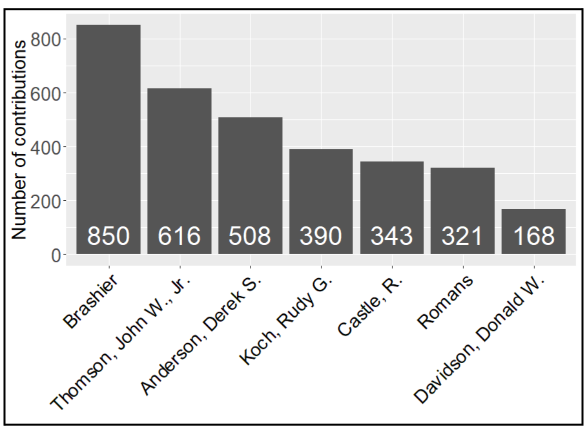


Figure 4 - Number of Contributions Made By The Top Seven Collectors.

Contributions across all seven top collectors are concentrated in northern Wisconsin, particularly in Douglas County where SUWS is located (see Figure 5). The ranges of two top collectors, Brashier (blue) and Romans (yellow) are notable for being nearly the same. The counties collected in and the number of specimens collected in the county tend to be similar between these two collectors, especially in counties farther from Douglas County.

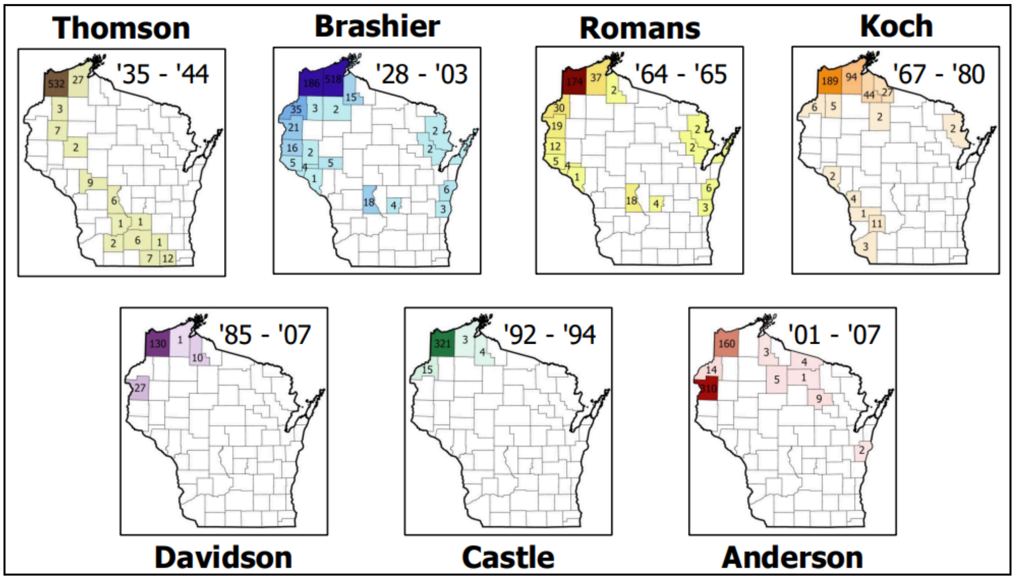


Figure 5 - Distribution in Wisconsin Counties Of The Top Seven Collectors. The maps are labeled with the last names of each collector, and the years of respective first and last contributions are noted.

The top seven collectors tend to contribute during different periods, and other collectors besides the top seven rarely make contributions outside these times (see Figure 6). For example, Brashier (blue) and Romans (yellow) were primarily active during a three year period from 1964 to 1966. In 1964, the highest year, 502 specimens representing 1,000 contributions were added to the collection – this single year was an approximately 40% increase in the physical SUWS collection at the time, and specimens from 1964 represent around 10% of the collection that is digitized. Not all collections contributed during this three-year period were contributed solely by Brashier (blue) and Romans (yellow); other collectors (pink) in addition to these two were also active in this period.

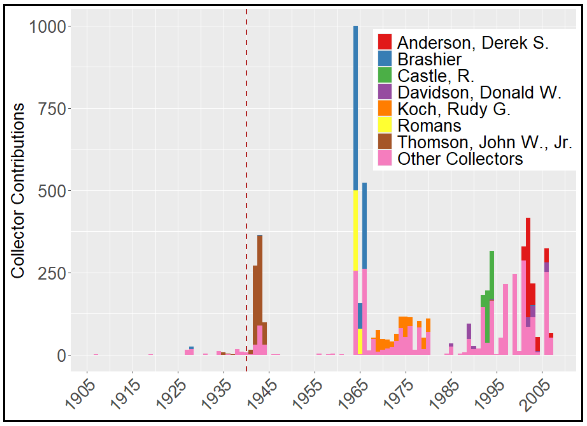


Figure 6 - SUWS Collector Contributions By Year. The red dashed line denotes the collection's official founding in 1940.

Co-collection was strong in SUWS. The comparison of contribution and specimen numbers discussed in a previous paragraph indicated it was common to have specimens with multiple contributors listed, and top collectors were no different in their practices. For example, Brashier contributed 526 specimens (61.9% of his 850 contributions) that were co-collections with collectors who were not the top seven of SUWS (see Figure 7). During his time contributing to SUWS, Brashier collected with 27 other people (see Figure 8). For 26 of them, their specimens co-collected with Brashier represent the majority of their contribution to SUWS; this group of 26 includes Romans, another top seven collector for SUWS, who contributed 321 specimens with Brashier.

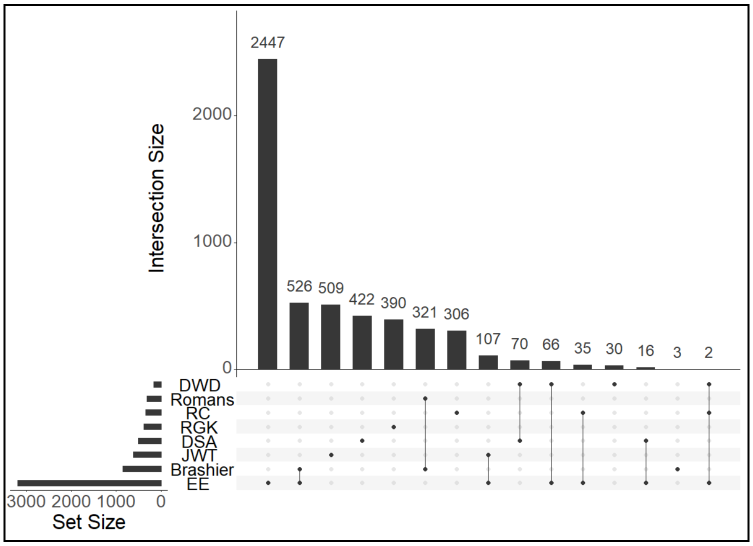


Figure 7 - Intersections Between Sets of SUWS Contributions. Set size indicates overall size of set, and the initials or last names identify the groups. The set EE refers to “Everyone Else”, or the set of 285 collectors that are not the top seven. Intersection size indicates by how many specimens the sets have, and the dots below indicate particular sets that are present in that number.

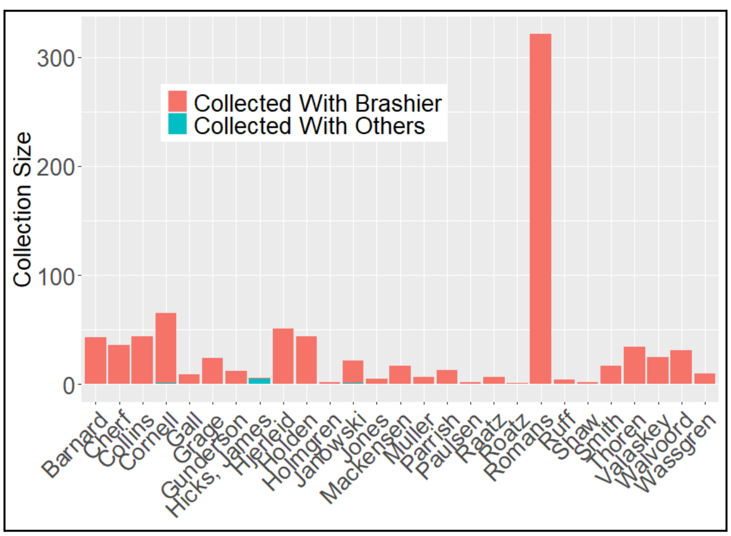


Figure 8 - Brashier’s 27 Co-Collectors.

## Discussion

Analysis of SUWS physical specimens and its database demonstrated how digitization and collection processes affect representation of specimens and associated collection phenomena. This kind of analysis that highlights collection biases present is particularly important for a collection that is incompletely digitized. Despite working with limited digitized records, I was able to identify a unique co-collection pattern with top contributors to SUWS that has historically driven collection growth.

Co-collection building the SUWS collection is likely not by chance, given that the herbarium is housed by an educational institution. Collection and co-collection specifically for classes may have driven herbarium growth. At least two of the top collectors, John W. Thomson, Jr. and Donald W. Davidson, were professors teaching classes during the respective periods they were making contributions to SUWS. In the raw data from WisFlora, some collectors were listed as 'Team 1' or with a similar epithet and this indicates student lab groups were assigned to collect specimens. When I took Plant Taxonomy from the University of Wisconsin - Superior in 2024, my final project, described by my professor as in keeping with how the course had been taught for a long time, was to go out, identify plants, record information, and press as if I were making a contribution to SUWS.

Future projects using digitized SUWS specimen data should further investigate the role of collectors during their time contributing to the collection to confirm the strong co-collection pattern described here. Further work with the collection and its database could also look at how digitizing specimens alters collection biases, or could compare SUWS to other regional herbaria such as the Olga Lakela Herbarium at the University of Minnesota - Duluth or the small collection housed by the College of St. Scholastica.

## References


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